

Cash Got Your Tongue? CEO Language Uncertainty Around Undisclosed Cash Acquisitions

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June 2026

1 Conceptual Framework and Empirical Strategy

1.1 Conceptual Framework

A firm that has privately committed to an acquisition but not yet announced it occupies a distinct *disclosure state*. A pending acquisition that is material to the acquirer constitutes material nonpublic information: information about preliminary merger negotiations can be material well before any definitive agreement, with materiality determined by balancing the probability that the transaction is consummated against its magnitude to the firm (Basic Inc. v. Levinson, 1988). Securities law imposes no general duty to disclose such confidential negotiations, so silence is permitted; but once a firm speaks it may not mislead, since an untrue or half-true statement is unlawful (Rule 10b-5, 2014; Basic Inc. v. Levinson, 1988). The quarterly earnings call is a recurring event that managers nonetheless hold, so the executive faces live questioning while unable to address the one material development and unable to deny it. Disclosure theory tells us such a withholding state is not informationally empty: an informed manager may rationally withhold private information when disclosure is costly, sustaining a threshold below which information is withheld (Verrecchia, 1983), and non-disclosure persists in equilibrium precisely because outside investors cannot distinguish a manager who is uninformed from one who is informed but silent (Dye, 1985). This withholding bind—material information held back while the firm keeps fielding questions—is the organizing primitive of our analysis, which we read in descriptive, correlational terms throughout.

The venue in which this bind becomes observable is the quarterly earnings conference call, which pairs a scripted managerial presentation with an unscripted question-and-answer session in which analysts question management directly. The two segments are not equally revealing: both are incrementally informative relative to the accompanying press release, and the discussion carries information content beyond the presentation (Matsumoto et al., 2011)—making the unscripted exchange, which managers cannot fully prepare in advance, the natural place to observe a constrained executive under questioning. That language can be read quantitatively: because general-purpose word lists misclassify financial text—nearly three-quarters of the Harvard dictionary’s negative-word counts are attributable to words that are typically not negative in a financial context—Loughran and McDonald (2011) construct dictionaries specific to financial disclosure, among them an uncertainty list of 285 words organized around imprecision rather than risk. We adopt this established apparatus—the presentation/Q&A split and the finance-specific dictionary, including its uncertainty list—to read the language of the call.

*I am deeply grateful to my co-advisors, Dr. Ali Akyol and Dr. Harshit Rajaiya, whose guidance, rigorous feedback, and steady support shaped every part of this thesis. All remaining errors are my own.

Within this venue the bind has an observable consequence. Because the pending acquisition is material and nonpublic, the executive can neither address the analysts' questions that bear on it nor deny that anything is afoot, so a direct answer is unavailable. Such non-answers are common, and they are not informationally empty: in roughly six of ten calls managers withhold requested information, leaving questions unanswered, and, in the words of that literature, silence speaks (Hollander et al., 2010). But silence is not a costless refuge—precisely because it speaks—and the setting is an unscripted exchange rather than prepared remarks. Non-disclosure therefore need not surface as outright silence; it can surface in the words themselves, as answers that grow hedged, qualified, and non-committal—elevated uncertainty language in the Q&A. We read this language, rather than the incidence of silence, because the constraint binds while the executive keeps speaking. The resulting pattern is keyed to the disclosure state: the bind predicts that uncertainty language is elevated while the acquisition remains undisclosed and recedes once announcement makes it public—a directional prediction we test rather than assume. This anticipatory timing—present before the announcement, absent after—is the first of the two dimensions we develop.

The anticipatory pattern just described, however, is not the only thing moving uncertainty language on a call. How uncertain a CEO sounds is partly a durable personal trait: some executives simply speak more cautiously than others, quarter after quarter. Bertrand and Schoar (2003) show that managers carry such persistent, individual styles—manager fixed effects—that durably shape the policies of the firms they run, with a manager's imprint in one firm strongly predicting his imprint in the next across investment, financial, and organizational policy alike. Dzielinski et al. (2021) construct such a decomposition for the language of earnings calls, separating a CEO's use of uncertainty words into a persistent, manager-specific style component and a time-varying, call-level residual. The implication for our purpose is a locating one. A persistent style is a fixed trait of the speaker, not a signal that tracks a particular, still-undisclosed transaction; an anticipatory component, by contrast, must emerge while a given deal is pending and subside once it is announced—it must vary from call to call within a single executive's tenure. If such a component exists, it therefore cannot reside in the persistent style; it can only live in the call-varying residual that remains once persistent style is netted out. This locates where an anticipatory signal must be—it does not claim the residual is that signal, which only the research design that follows can isolate.

A second dimension concerns not the timing of the signal but the deals in which it should concentrate. Cash and stock acquisitions share the identical disclosure setting—both are material, both are withheld—yet differ in one design-relevant respect: a cash purchase draws on an accumulated cash position, whereas a stock exchange need not. Harford (1999) documents the underlying fact: firms accumulate cash reserves above a baseline model of normal holdings, and cash-rich firms are more likely to make acquisitions. That accumulation is a by-product of retained free cash flow rather than saving toward a particular deal, so we take from it only that a cash bid draws on a pre-existing cash position—not that firms stockpile in order to fund planned acquisitions. This makes the stock deal a natural placebo: the same withholding bind, minus the cash commitment. A signal that concentrates in cash deals rather than stock deals is therefore informative on its own terms, even though why the two should differ is a question we leave open. The two dimensions also run on different clocks: the anticipatory signal tracks the undisclosed information and resolves once the deal is announced, while the cash commitment serves the purchase itself and—mechanically—persists until it is paid at completion. The contrast is one of timing: the uncertainty is resolved by the announcement, the cash is not.

This paper's nearest neighbors read deal-related language along dimensions other than ours. Thewissen et al. (2024) show that acquirers manage the tone of their earnings press releases: stock bidders inflate disclosure tone by about 15 percent in the year before a stock-for-stock announcement, lifting their share price to reduce the cost of the shares they must issue. Ragozzino and Reuer (2024) find a parallel pattern in the spoken record—on the earnings calls of firms active in M&A, the volume of corporate-strategy vocabulary rises, with analysts using such terms nearly 9 percent more, and executives 7.2 percent more, than at inactive firms. Both study language that is deliberately deployed or topic-driven, and both concern tone or subject

matter rather than expressed uncertainty. Separately, Keown and Pinkerton (1981) document that prices often move ahead of a deal: roughly half of the market’s reaction to a merger occurs before its first public announcement. That anticipatory signal, however, lives in prices, not in language. The cell these literatures leave empty is the one this paper occupies: uncertainty language in the unscripted Q&A, in the anticipatory window before the deal is public, read for what it may reveal about a deal the firm is withholding rather than as language crafted to move the market. To our knowledge, no prior work reads this; we offer it as a positioning claim about where the contribution sits, not as a tested mechanism.

Two readings of this prediction are observationally equivalent, and the framework does not require us to choose between them. Under the first—compliance-constrained disclosure—the executive is unable to confirm the pending deal and barred from denying it, is left to answer around it, and uncertainty rises as a by-product of the bind. Under the second—strategic silence—the reticence is deliberate: the executive chooses to stay vague about a development that is not yet public. Both would produce the same observable signature: elevated uncertainty language in the Q&A while the deal is undisclosed, resolving at announcement. Our design cannot distinguish them, and we do not try. What we characterize is therefore a correlational pattern keyed to the disclosure state—an anticipatory, cash-concentrated component of CEO uncertainty language that tracks a deal’s passage from private to public—and not a mechanism. We identify no causal channel, establish no intent, and read the relationship descriptively throughout. It is in this deliberately bounded register that the empirical strategy of the following sections puts the pattern to the test.

1.2 Hypothesis Development

The preceding section isolates two dimensions of the pattern we expect. The first is an *anticipatory* dimension, in which CEO uncertainty language is elevated while an acquisition remains undisclosed and recedes once it is announced. The second is a *cash-concentrated* dimension, in which that elevation is stronger for cash acquisitions than for stock. This section turns those dimensions into formal, falsifiable predictions about *where* and *when* the signal appears in a CEO’s unscripted answers, not why it appears, a mechanism the framework leaves open. The object of each prediction is the call-varying residual of CEO uncertainty language in the Q&A, the component that remains once a CEO’s persistent speaking style is netted out (defined formally in the next section), and every prediction is read in descriptive, correlational terms. Timing is measured in event time: e counts calendar quarters relative to a firm’s first acquisition announcement, so that $e = -1$ denotes the call in the quarter immediately before it, and the focal pre-announcement indicator is $\text{PreAnnounceQtr} = \mathbf{1}[e = -1]$, defined for a firm’s first acquisition financed at least half in cash. To state the predictions compactly, we write θ_e for the shift in residual Q&A uncertainty at event position e relative to a firm’s own baseline quarters (for cash acquirers, unless a deal-type superscript such as stock indicates otherwise), and κ_e for the analogous shift in its cash-to-assets ratio; each hypothesis below is a sign restriction on these shifts, which the designs of Section 2.4 estimate. The three hypotheses, H1, H1a, and H1b, are stated both informally and formally and then confronted with the data.

The first hypothesis concerns timing. While the deal is withheld, the executive faces questions that bear on it but can neither address nor deny it, so residual uncertainty should run high in the quarter before the announcement and subside once the deal is public. That this is not mechanical is clear from the nearest evidence, which runs the other way. Thewissen et al. (2024) document that acquirers manage the tone of their disclosure ahead of stock-for-stock deals, a deliberate adjustment; H1 predicts a different register in a different setting, not tone managed to move the market but uncertainty language that surfaces in a cash acquirer’s answers while the deal is withheld. Consistent with the bounded reading established earlier, the hypothesis takes no stance on whether that surfacing is compliance-constrained or strategically chosen; it asserts only that the language moves, and when.

H1 (Timing).

Informally: residual Q&A uncertainty is higher in a cash acquirer’s pre-announcement quarter

than in that same firm’s other quarters.

Formally: $\theta_{-1} > 0$.

The second hypothesis sharpens the first by asking where the elevation should concentrate. Cash and stock acquisitions share the disclosure setting, both material and both withheld, yet differ in one respect the design exploits: a cash purchase draws on an accumulated cash position, whereas a stock exchange need not. As developed in the framework, this makes the stock deal a natural placebo, the same withholding bind without the cash commitment, against which a cash acquirer’s run-up can be read. The claim is one of concentration rather than strict specificity, and it concerns the run-up in language alone; H1a makes no claim that the underlying cash accumulation is itself peculiar to cash acquirers, and it asserts no cause.

H1a (Concentration).

Informally: the pre-announcement elevation in residual uncertainty is larger for cash acquirers than for stock acquirers.

Formally: $\theta_{-1}^{\text{cash}} > \theta_{-1}^{\text{stock}}$.

The third hypothesis concerns resolution. The residual uncertainty and the acquirer’s cash run on different clocks. The uncertainty tracks the undisclosed information and should resolve at announcement, receding once the deal is public even before it closes, whereas the cash that funds the purchase persists on the balance sheet until it is paid at completion. The most discriminating window is the gap between announcement and completion, when the deal is announced but not yet closed: there the language should already have returned to baseline while the cash still sits unspent. That the cash leg is largely mechanical, since cash leaves only when the deal is paid, is by design; H1b predicts the divergence of the two paths, not the level of either.

H1b (Resolution).

Informally: residual uncertainty returns to baseline as soon as the deal is announced, even before it closes, whereas the cash that funds it remains in place until completion.

Formally: $\theta_{-1} > 0$ and $\theta_{\text{gap}} = 0$, while $\kappa_{\text{gap}} > 0$.

One alternative to this anticipatory reading should be flagged. The pre-announcement rise might reflect not the disclosure bind but analyst scrutiny: analysts spending more of the call on cash and liquidity, mechanically drawing out hedged answers. This is a plausible alternative driver of the run-up, distinct from the framework’s two open readings rather than a third version of them, and the design treats it as an identification threat to confront. Its construction and motivation are developed in Section 2.5, and it is put directly to the data as a side analysis in Section 4.1. Flagging it here records that the run-up’s most immediate competing explanation is tested, with the verdict reported there; confronting it bounds the alternatives without establishing the disclosure mechanism, which the framework leaves open.

1.3 Estimation of the Main Variable

The dependent variable is built from the language of the call. Following the apparatus adopted in Section 2.1, we measure uncertainty with the Loughran and McDonald finance-specific uncertainty word list, applied as a word share. For a CEO’s answers in the question-and-answer session, the raw measure is the share of uncertainty words among all the words the CEO speaks when answering analysts, which is Equation (2) of Dzielinski et al. (2021):

$$\text{UncAnsCEO} = \frac{\text{UnctWordsAnsCEO}}{\text{WordsAnsCEO}}. \quad (1)$$

This raw share is the quantity the decomposition below operates on. The same word-share form is computed for the scripted presentation, UncPreCEO , and for the analysts’ own questions, UncQue ; these enter the decomposition as controls, not as the dependent variable. How the transcripts are tokenized and the words

matched is a construction detail we defer to Section 3.1; here the measure is taken as given, and the question is which part of it can carry an anticipatory signal.

An anticipatory, deal-tracking signal must vary from call to call within a single CEO’s tenure, so, as established in Section 2.1, it can reside only in the call-varying part of this measure and not in the CEO’s persistent speaking style. To separate the two, we adopt the decomposition in Equation (4) of Dzielinski et al. (2021), estimated on our sample:

$$\text{UncAnsCEO}_{i,t} = \alpha + \gamma_i \text{CEO}_{i,t} + \beta_s \text{Speech}_{s,i,t} + \beta_k \text{FirmChars}_{k,i,t} + \text{Year}_t + \varepsilon_{i,t}, \quad (2)$$

where $\text{CEO}_{i,t}$ is an indicator for manager i being CEO in quarter t , $\text{Speech}_{s,i,t} = \{\text{UncPreCEO}, \text{UncQue}, \text{NegCall}\}$ collects other linguistic features of the same call, $\text{FirmChars}_{k,i,t} = \{\text{SurpDec}, \text{EPSgrowth}, \text{StockRet}, \text{MarketRet}\}$ collects contemporaneous performance, and Year_t absorbs common annual conditions. The persistent component is the CEO fixed effect, summarized as $\text{ClarityCEO}_i = -\gamma_i$, and the object of interest is its residual term, $\text{UncResCEO} = \varepsilon_{i,t}$, which we estimate on our data. Because UncResCEO is taken net of the persistent style and of these observable call-level controls, it is narrower than a style-only residual: a sharper version of the call-level component located in Section 2.1, not a different object. The authors themselves construct this residual and use it as an explanatory variable in their own market-response model (their Equation (5)); our contribution is not the measure but the anticipatory, disclosure-state reading we give it around undisclosed deals. Finally, because the residual is net of the scripted presentation, UncPreCEO , it isolates the unscripted component, the place where the anticipation of an undisclosed deal would surface; should some of that anticipation also reach the prepared remarks, the residual understates it and so reads as a conservative floor. By the same construction it is net of analyst-question uncertainty, UncQue , and of call negativity, NegCall , a property Section 2.5 relies on when it separates the signal from analyst scrutiny.

One feature of this construction deserves note. Because UncResCEO is itself the residual from a first-stage regression and then serves as the dependent variable in the analyses that follow, the design is a standard two-step one, and conventional second-stage standard errors can be understated unless a two-step correction is applied. We flag the property rather than treat it as a defect: it is generic to two-step estimation, Dzielinski et al. themselves use the residual as a regressor, and where it matters a bootstrap that re-estimates both stages supplies a remedy. The deeper question, whether the residual reflects genuine uncertainty rather than merely the analyst scrutiny a call attracts, is not settled by construction; we treat it as an empirical matter and take it up in Section 2.5.

1.4 Methodology and Empirical Design

Each hypothesis of Section 2.2 maps to an estimating equation, and all three share a common panel frame: a within-firm, two-way fixed-effects regression of an outcome on event-time treatment indicators, estimated on quarterly firm observations with standard errors clustered by firm. The main design tests H1. Writing Y_{it} for the outcome of firm i in quarter t ,

$$Y_{it} = \beta \text{PreAnnounceQtr}_{it} + \gamma' X_{it} + \phi_i + \tau_t + \varepsilon_{it}, \quad (3)$$

where X_{it} is a vector of firm controls (leverage, log assets, Tobin’s Q , return on assets, capital expenditure, a dividend indicator, and scaled cash flow from operations), ϕ_i is a firm fixed effect, and τ_t is a calendar year-quarter fixed effect. The treatment $\text{PreAnnounceQtr}_{it} = \mathbf{1}[e = -1]$ marks the call in the quarter before the firm’s first cash acquisition; post-announcement quarters ($e \geq 0$) are dropped, so the design reads the run-up and not the aftermath, while firms with no cash acquisition, for which e is undefined, anchor the calendar-time effects. The coefficient β is the estimand: it estimates the pre-announcement shift θ_{-1} of Section 2.2, a within-firm contrast between a treated firm’s pre-announcement quarter and its own ordinary quarters. We read it descriptively, as an anticipation contrast, and make no causal or parallel-trends claim.

H1b requires tracing the outcome across event time rather than at a single point, so the main design generalizes to an event study that replaces the single indicator with a set of event-time bins:

$$Y_{it} = \sum_b \beta_b \text{Bin}_{b,it} + \gamma' X_{it} + \phi_i + \tau_t + \varepsilon_{it}, \quad (4)$$

with four bins: PRE2 ($e = -2$), PRE1 ($e = -1$), GAP (announced but not yet completed), and POST (completed). The omitted baseline is the set of quarters at $e \leq -3$ together with never-acquirers; the post-announcement window is capped at four quarters and truncated at a firm's next announcement. The discriminating window is the gap. If residual uncertainty has resolved by announcement, its coefficient should fall to zero at GAP, when the deal is public but the cash has not yet left, so a Wald test of $\beta_{\text{PRE1}} - \beta_{\text{GAP}} > 0$ carries the empirical content of H1b; the same specification, estimated for the cash ratio in place of uncertainty, traces the persistence of the cash on the other clock, a persistence that is largely mechanical, since the cash is paid only at completion and so sits on the books through the gap. PRE2 is a pre-trend check, expected to be near zero. The bins are estimated for cash acquirers and, as a placebo, for stock acquirers, and every coefficient and Wald contrast is reported two-tailed.

H1a is a claim about the difference between cash and stock acquirers, so it is tested in a single pooled specification carrying both treatments at once:

$$Y_{it} = \beta_c \text{PreAnnCash}_{it} + \beta_s \text{PreAnnStock}_{it} + \gamma' X_{it} + \phi_i + \tau_t + \varepsilon_{it}, \quad (5)$$

where PreAnnCash_{it} and PreAnnStock_{it} mark the quarter before a firm's first cash and first stock acquisition. The sample keeps only the quarters before a firm's first deal of either type, plus firms that make neither, which form the fixed-effects baseline. The formal cash-specificity test is the single linear restriction $\beta_c - \beta_s > 0$, evaluated by a Wald test; this difference, and not a comparison of one significant coefficient against one insignificant one, is what estimates $\theta_{-1}^{\text{cash}} - \theta_{-1}^{\text{stock}}$ in H1a. When the outcome is the cash ratio rather than residual uncertainty, the specification adds the cash ratio's own one-quarter lag, so the pre-announcement coefficient tracks the change in cash rather than its level; the residual outcome takes no such lag.

The identifying assumptions are descriptive and common to the three designs. Identification is within firm, through the firm fixed effects, and within calendar time, through the year-quarter fixed effects; the estimand in each case is a pre-announcement mean shift relative to a firm's own ordinary quarters and to untreated firms. The PRE2 bin is a validity device rather than an identification proof: a pre-trend near zero is consistent with the reading, but we claim no parallel-trends result and no causal effect. The mapping from design to hypothesis is one to one: the single-indicator regression tests H1, the event study tests H1b, and the pooled interacted regression tests H1a. The control pool itself differs across designs: in the single-indicator and event-study regressions a firm remains in the baseline until its first cash acquisition, so stock-only acquirers sit among the controls, whereas the pooled regression restricts the baseline to firms that make neither a cash nor a stock acquisition. Two distinct control layers operate and should not be conflated. The residual outcome is already net of a first set of controls, the linguistic and performance variables of the decomposition in Section 2.3, estimated with CEO and year fixed effects; the designs here add a second set, the firm-financial controls X_{it} , with firm and finer year-quarter fixed effects. The two layers serve different purposes and are estimated at different stages.

Several choices that do not belong to a single equation are recorded here. On inference, standard errors are clustered by firm throughout, while some validity tables cluster two ways, by firm and by calendar quarter; focal treatment tests are reported two-tailed, the convention we adopt for honesty even where a directional prediction would license a one-tailed test. On contamination, deal timing is endogenous, and because only a firm's first deal sets the event clock, later deals can pollute the baseline; we mitigate this by truncating each firm's post-window at its second announcement, but the never-acquirer baseline and the first-deal focus remain assumptions, not identification. On functional form, the outcomes are levels or word shares rather

than counts, the cash-ratio regressions carry a partial-adjustment lag because cash is sticky whereas the residual regressions do not, and, because the residual is a generated regressand from Section 2.3, the two-step standard-error caveat carries over to every design here. On selection, the residual outcome exists only for CEOs with at least five calls, in the non-financial, non-utility sample where the decomposition is run, so the uncertainty regressions describe a larger, more-covered set of firms than the cash-ratio regressions, a limit on external validity we disclose. The firm-financial controls follow the standard cash-determinants regression, read mechanically, with cash treated as a by-product of retained free cash flow rather than a war chest built for a deal.

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